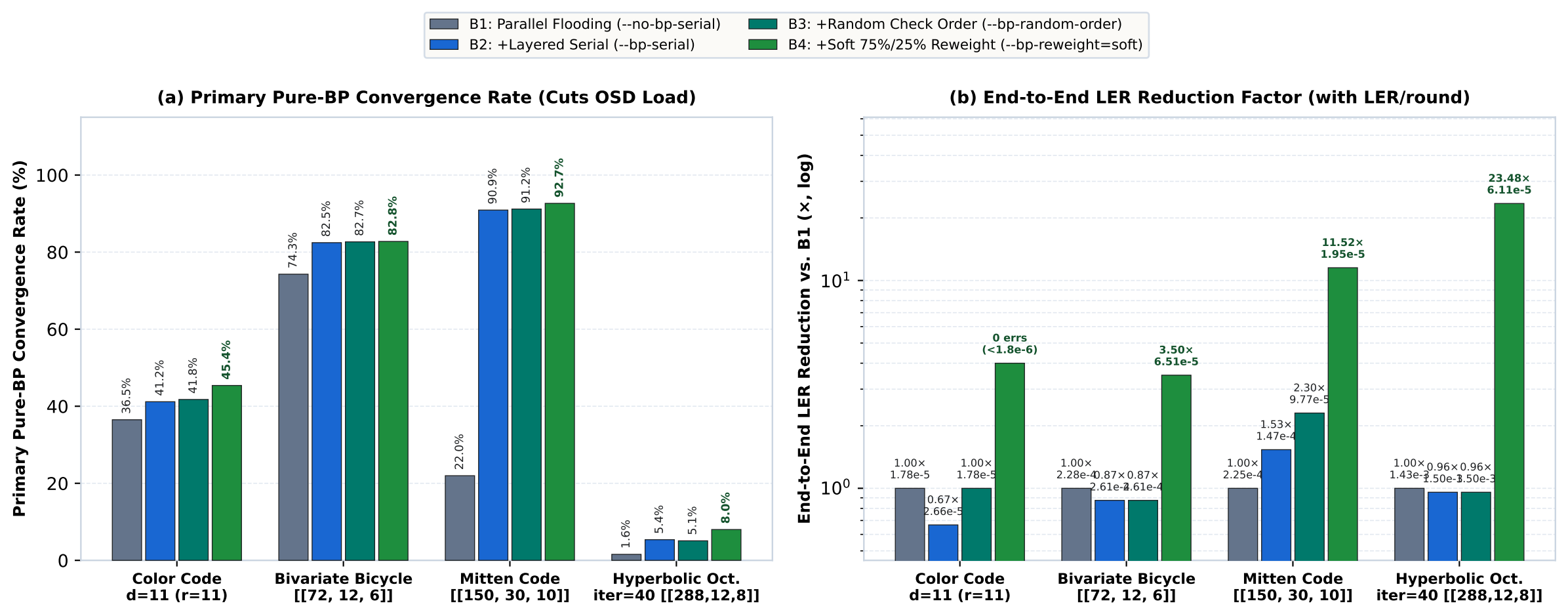
B1: Parallel Flooding (--no-bp-serial)
B2: +Layered Serial (--bp-serial)
B3: +Random Check Order (--bp-random-order)
B4: +Soft 75%/25% Reweight (--bp-reweight=soft)
(a) Primary Pure-BP Convergence Rate (Cuts OSD Load)
Primary Pure-BP Convergence Rate (%)
36.5%
41.2%
41.8%
45.4%
74.3%
82.5%
82.7%
82.8%
22.0%
90.9%
91.2%
92.7%
1.6%
5.4%
5.1%
8.0%
Color Code d=11 (r=11)
Bivariate Bicycle [[72, 12, 6]]
Mitten Code [[150, 30, 10]]
Hyperbolic Oct. iter=40 [[288,12,8]]
(b) End-to-End LER Reduction Factor (with LER/round)
End-to-End LER Reduction vs. B1 (×, log)
1.00× 1.78e-5
0.67× 2.66e-5
1.00× 1.78e-5
0 errs (<1.8e-6)
1.00× 2.28e-4
0.87× 2.61e-4
0.87× 2.61e-4
3.50× 6.51e-5
1.00× 2.25e-4
1.53× 1.47e-4
2.30× 9.77e-5
11.52× 1.95e-5
1.00× 1.43e-3
0.96× 1.50e-3
0.96× 1.50e-3
23.48× 6.11e-5
Color Code d=11 (r=11)
Bivariate Bicycle [[72, 12, 6]]
Mitten Code [[150, 30, 10]]
Hyperbolic Oct. iter=40 [[288,12,8]]